206 Work, Energy and Power

2.6.1 · 2.6.2 · 2.6.3 Work done by a force

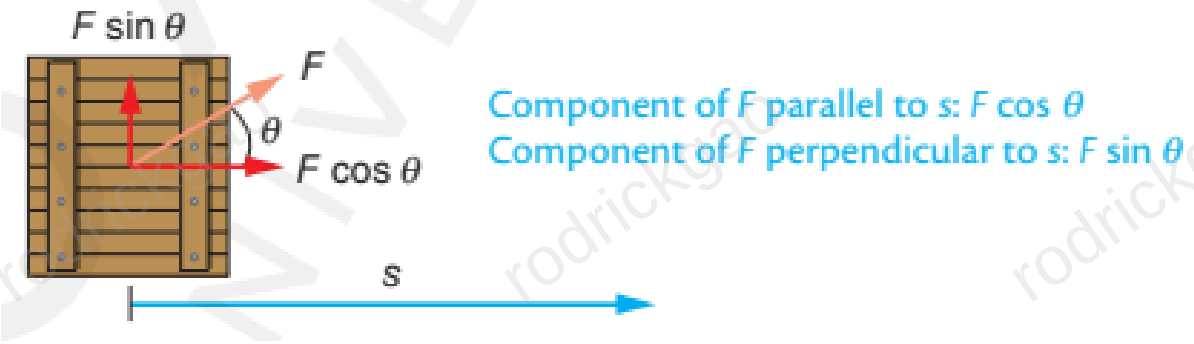
A force transfers mechanical energy through displacement, but only its component along the displacement does work. A component in the same direction does positive work; an opposing one does negative work; a perpendicular force does none. Force without displacement also does no such mechanical work. The constant-force formula uses the angle between force and displacement.

$$W = F_{\parallel} s$$

$$W = Fs\cos\theta$$

W: work (J); F∥: constant force component along displacement (N); s: displacement magnitude (m). A component opposite the displacement gives negative work.

W: work (J); F: constant force (N); s: displacement (m); θ: angle between force and displacement.



Only the force component along the displacement does work.

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2.6.4 Units of work and energy

A force of 1 N moving its application point 1 m in the force's direction does 1 J of work. Work is energy transfer, so both use joules. Although moment can also be expressed in N m, it describes turning effect rather than energy and is not normally expressed in joules.

$$1\ \mathrm{J} = 1\ \mathrm{N\,m}$$

One joule is the work of a one-newton force through one metre along the force. Torque also has unit N m, but is not energy.

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2.6.5 · 2.6.14 Kinetic energy

Translational kinetic energy is energy due to motion, proportional to mass and the square of speed. Doubling speed quadruples kinetic energy; changing direction alone while keeping speed fixed does not change it. Kinetic energy is a non-negative scalar. The classical expression applies at speeds much lower than the speed of light.

$$K = \frac{1}{2}mv^2$$

K (KE): translational kinetic energy (J); m: mass (kg); v: instantaneous speed (m s^{-1}). Non-relativistic motion.

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2.6.6 · 2.6.7 · 2.6.15 Gravitational potential energy and height

Near Earth's surface in an approximately uniform field, raising an object increases gravitational potential energy by its weight times its vertical height increase. The zero level is arbitrary but must remain consistent within a problem. Use vertical height, not slope length or path length. During descent, lost potential energy can become kinetic or other forms of energy.

$$U = mgh$$

$$\Delta U = mg\Delta h$$

U (PE): gravitational potential energy (J); m: kg; g: N kg⁻¹; h: height above the chosen zero level (m). Approximately uniform gravitational field.

ΔU: change in gravitational potential energy (J); m: kg; g: N kg⁻¹; Δh: final minus initial height (m). Constant g.

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2.6.8 Conservation of energy

Energy can be transferred between objects and converted between kinetic, potential, internal and other forms, but cannot be created or destroyed. Total energy remains constant in an isolated system. For a system exchanging energy with its surroundings, include all inputs and outputs. Friction reduces mechanical energy without violating conservation of total energy.

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2.6.9 Conservation of mechanical energy

Kinetic plus potential energy remains constant when only conservative forces do work and there is no other energy input or loss. Falling converts potential into kinetic energy; rising reverses this. If friction, air resistance or an engine does appreciable work, include that transfer rather than equating the initial and final mechanical energies directly.

$$K_1 + U_1 = K_2 + U_2$$

Mechanical energy conservation when only conservative forces do work. K: kinetic energy; U: potential energy (J).

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2.6.10 Mechanical energy lost against resistance

Motion against resistance converts some mechanical energy into internal or other forms. With no other input, initial mechanical energy equals final mechanical energy plus this loss. The loss is written as a positive amount, equal in magnitude to the negative work done by resistance. Total energy is not destroyed.

$$K_1 + U_1 = K_2 + U_2 + W_{\text{res}}$$

$W_{\text{res}} \geq 0$: energy transferred by work against resistance (J). No other energy input; K: kinetic energy; U: potential energy.

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2.6.11 External work and mechanical energy

Net non-conservative work changes the system's mechanical energy: positive work increases it and negative work decreases it. If gravity is already represented by gravitational potential energy, do not add gravitational work again. Define the system, energy reference level and forces represented by potential energy before writing the balance.

$$K_1 + U_1 + W_{\text{ext}} = K_2 + U_2$$

W_{ext}: signed work by external non-conservative forces (J). Do not count gravitational work again when gravitational potential energy is included.

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2.6.12 Power: the rate of energy transfer

Power measures energy transferred or converted per unit time. Average power is the total energy transferred divided by the elapsed time; 1 W means 1 J per second. Transferring the same energy in less time requires greater average power. If power varies, this average need not equal the instantaneous power at any chosen moment.

$$P_{\text{av}} = \frac{W}{\Delta t}$$

P_{av}: average power (W); W: work (J); Δt: elapsed time (s).

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2.6.13 · 2.6.16 Instantaneous power from force and velocity

The instantaneous power of a force is its component along the velocity multiplied by the instantaneous speed. Parallel force and velocity give their simple product; perpendicular force gives zero power and opposing force gives negative power. An average speed over an interval cannot generally replace the instantaneous speed when finding power at a particular moment.

$$P = \vec{F} \cdot \vec{v} = Fv\cos\theta$$

$$P = Fv(\theta = 0)$$

Instantaneous power P (W); F: force (N); v: velocity (m s^{-1}); θ: angle between them. The textbook's P = Fv assumes parallel force and velocity.

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